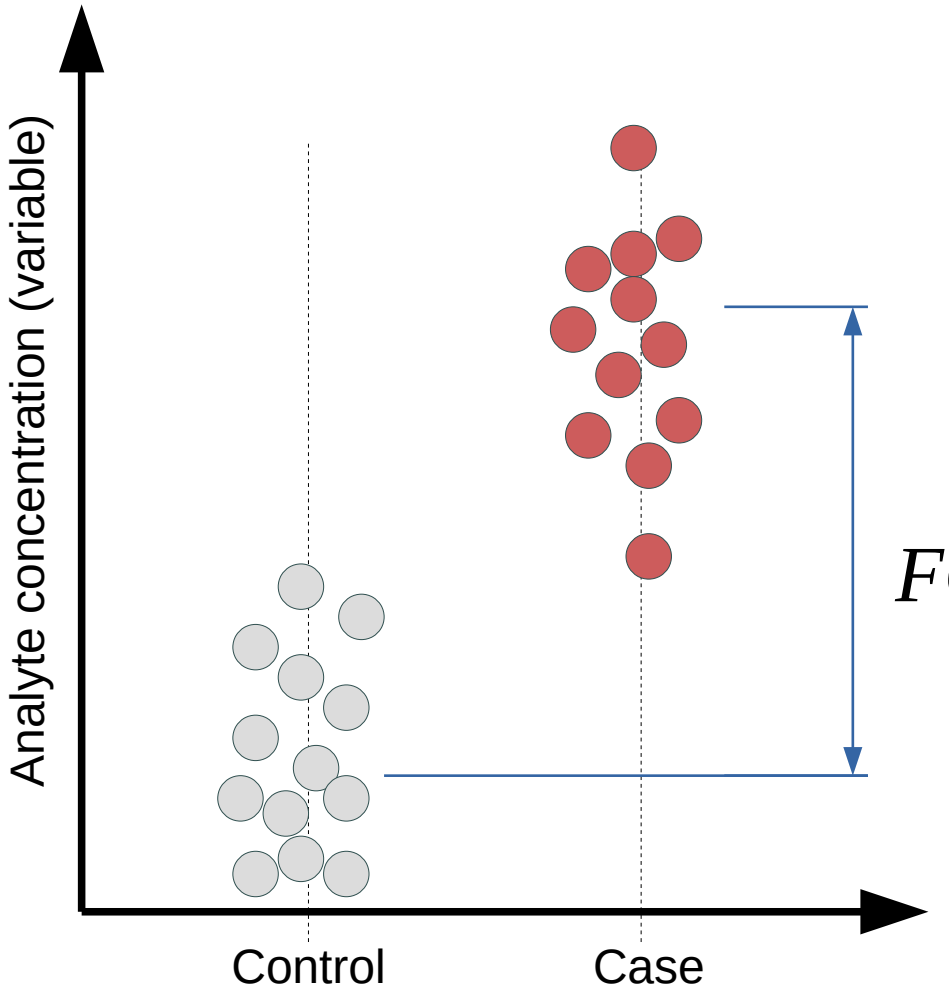
Data analysis and bio-statistics

Practice 4: Statistical Significance

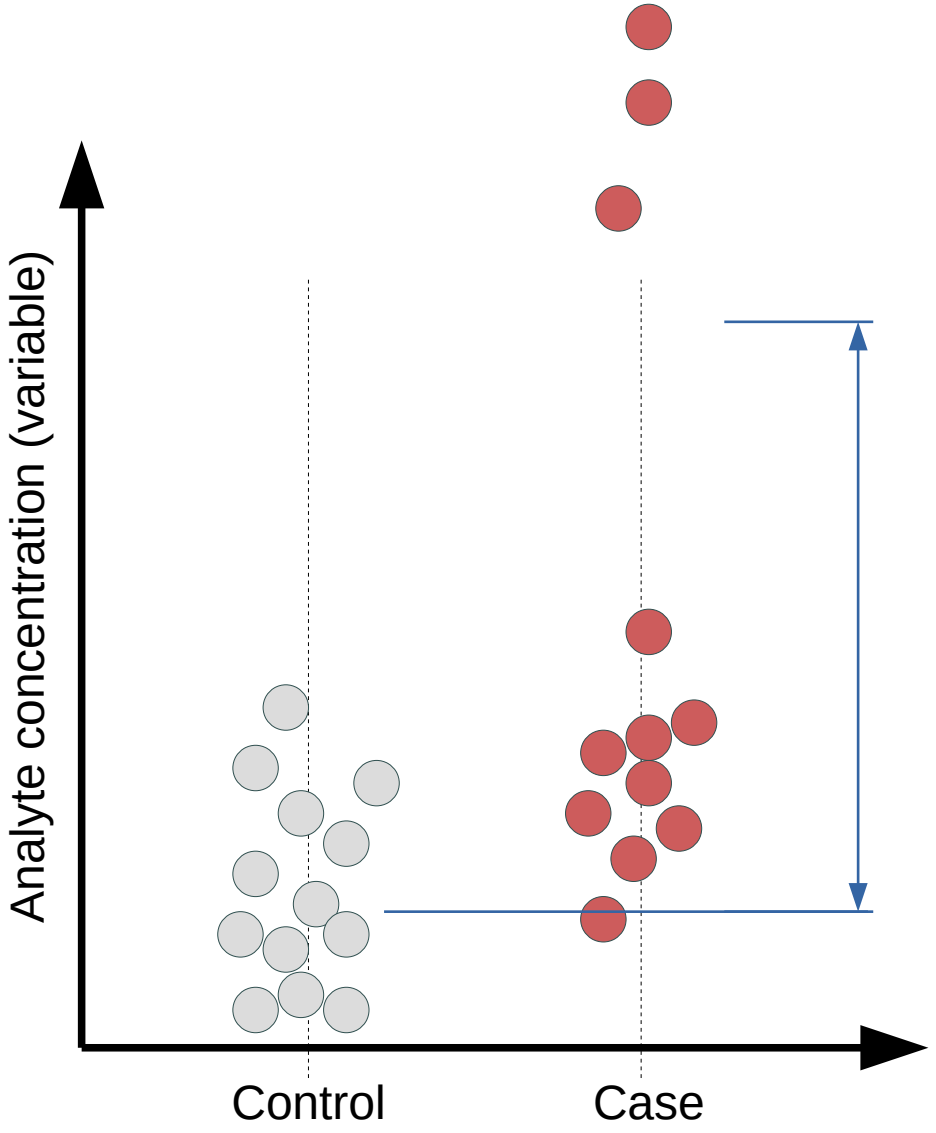
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Fold Change (FC)

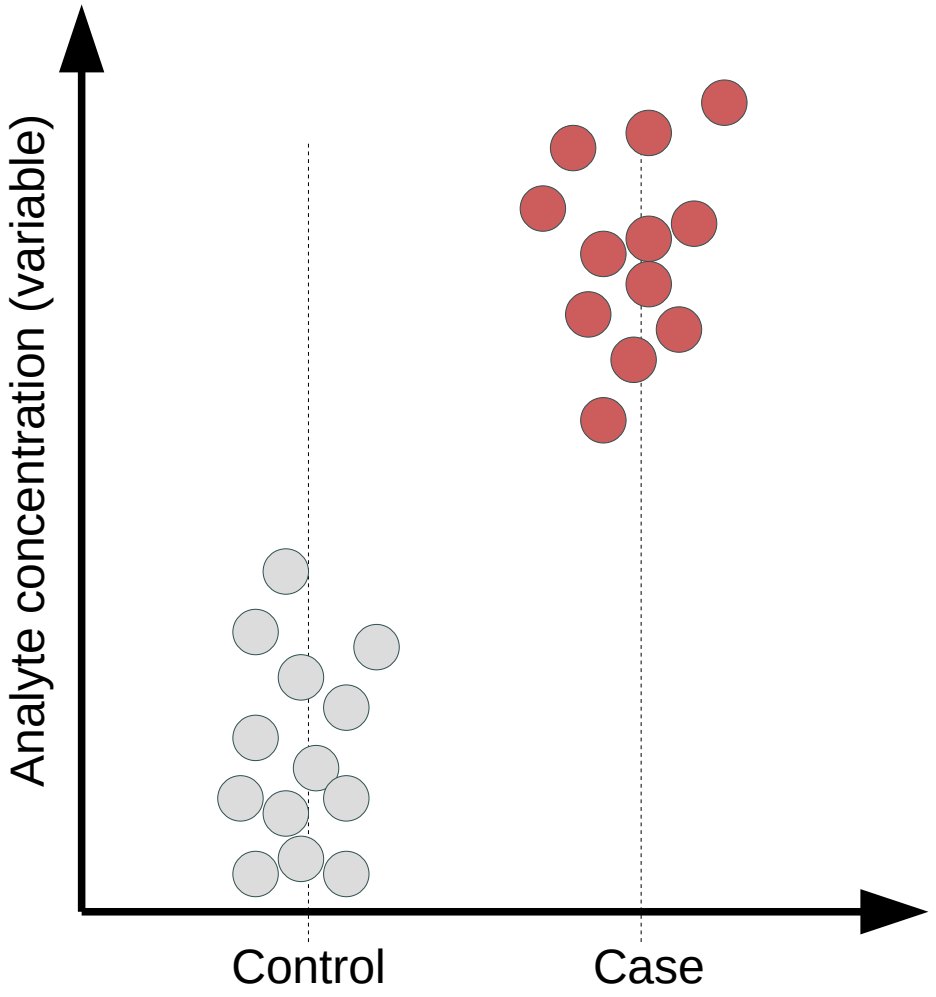


$$FC = \log_2\left(\frac{\overline{x}_{case}}{\overline{x}_{control}}\right)$$

Is Fold Change enough?

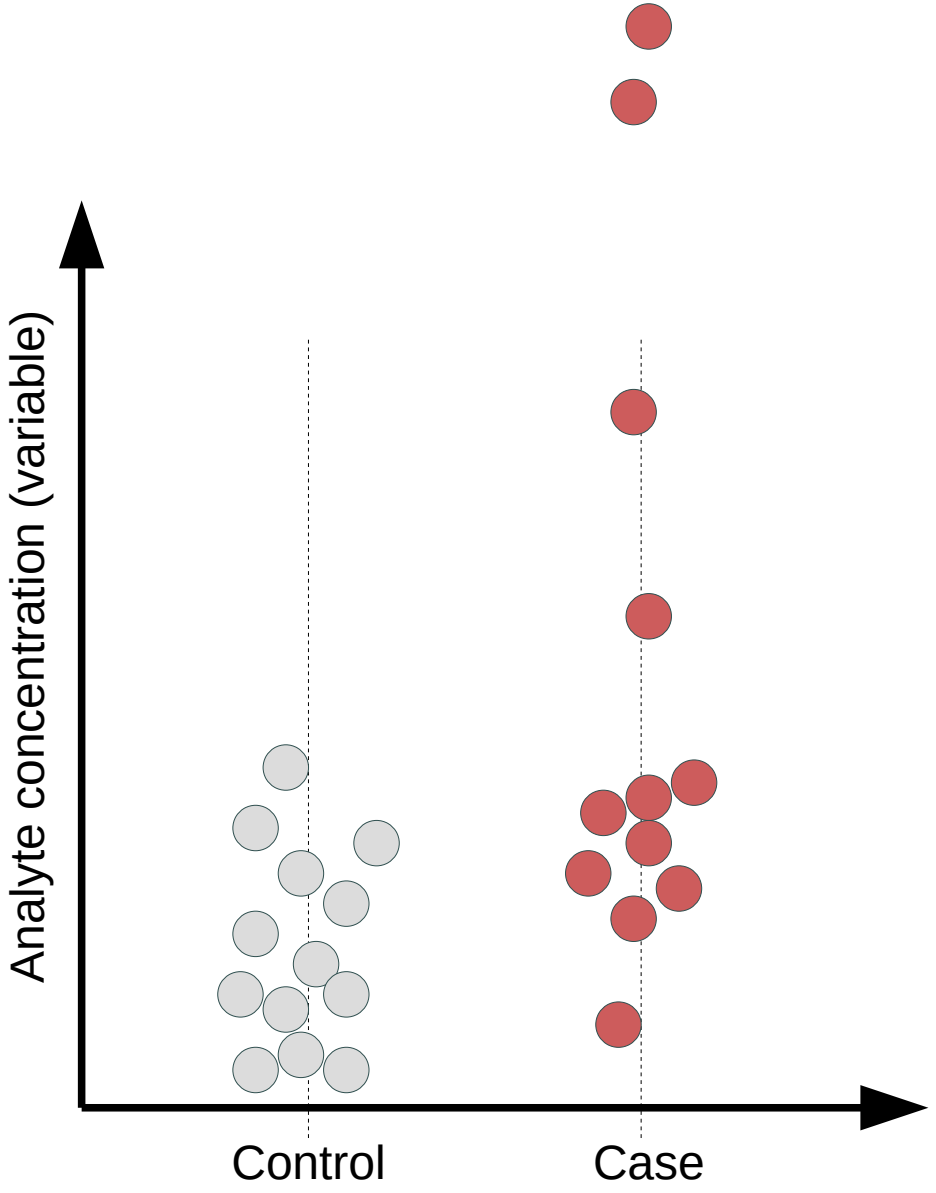


p.value: a significance measurement



FC → High
p.value → Low

p.value: a significance measurement



FC → High
p.value → High

p.value cut-off

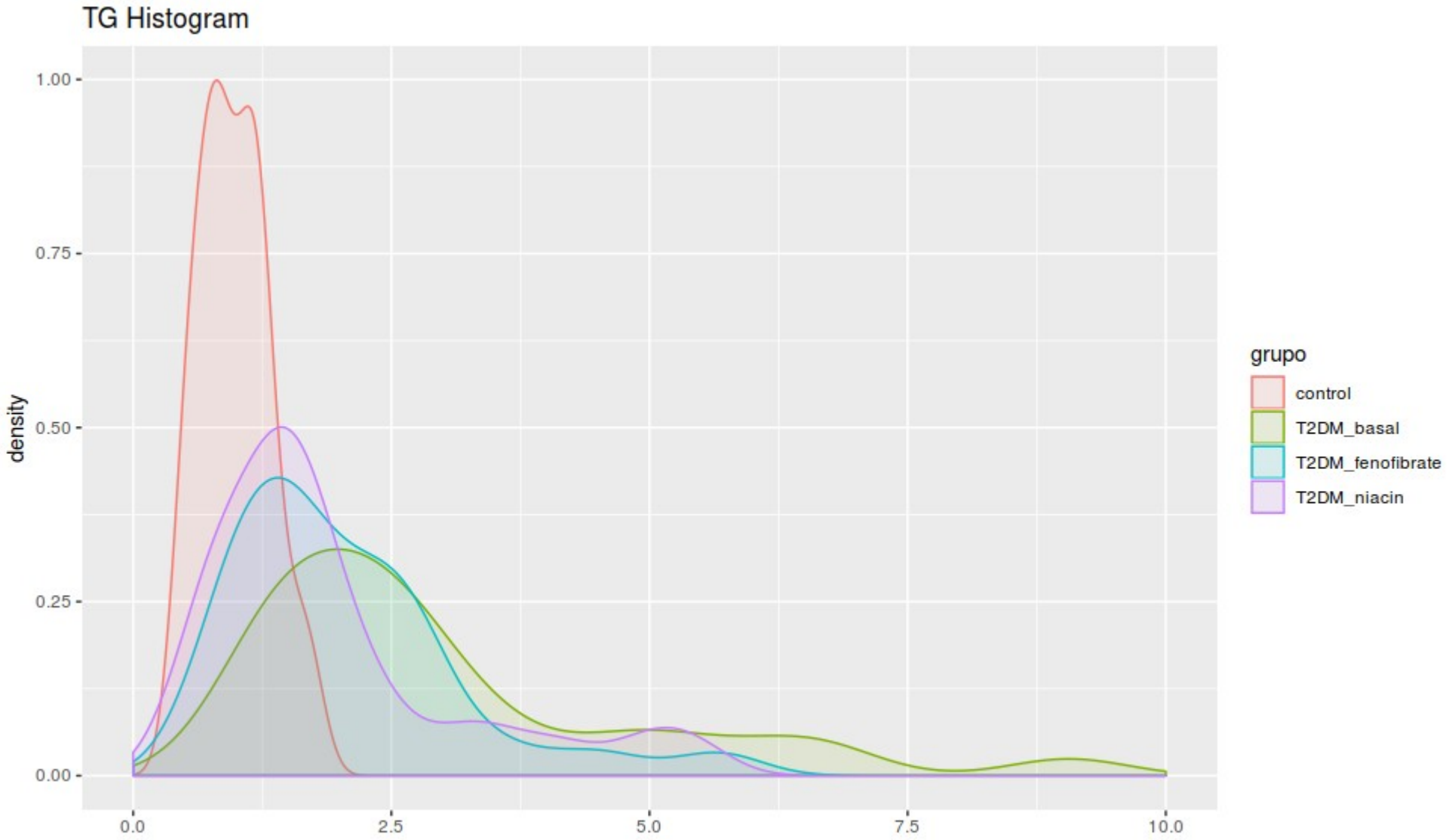
In bio-sciences it is accepted to consider a variable statistically significative when:

p.value < 0,05

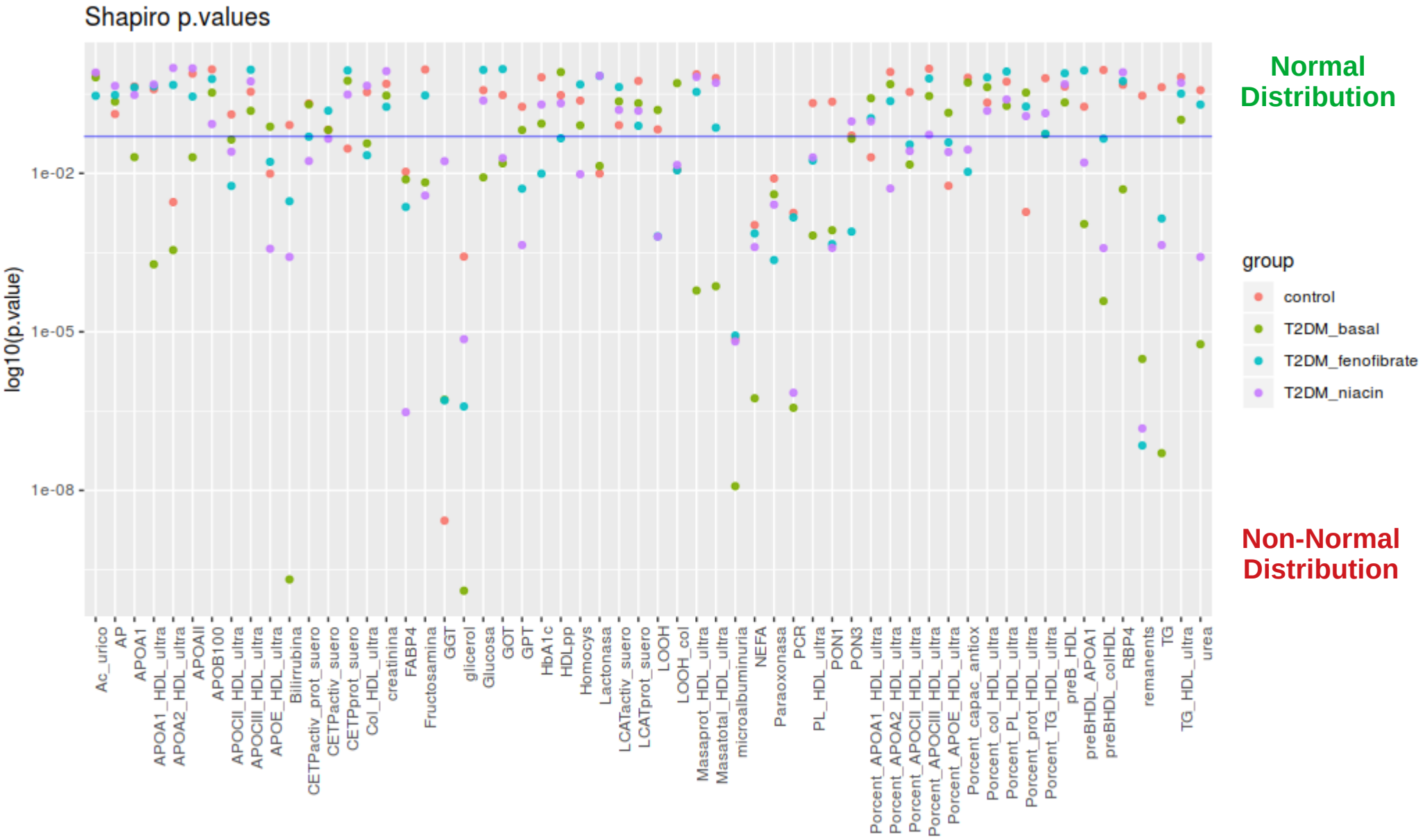
Methods to calculate the p.value

- Student T test
  - Parametric (assumes normal distribution)
  - Paired
- Mann–Whitney U test
  - Non-parametric
  - Paired
- Kruskal–Wallis H test
  - Non-parametric
  - Two or more independent samples

Has our data a normal distribution?

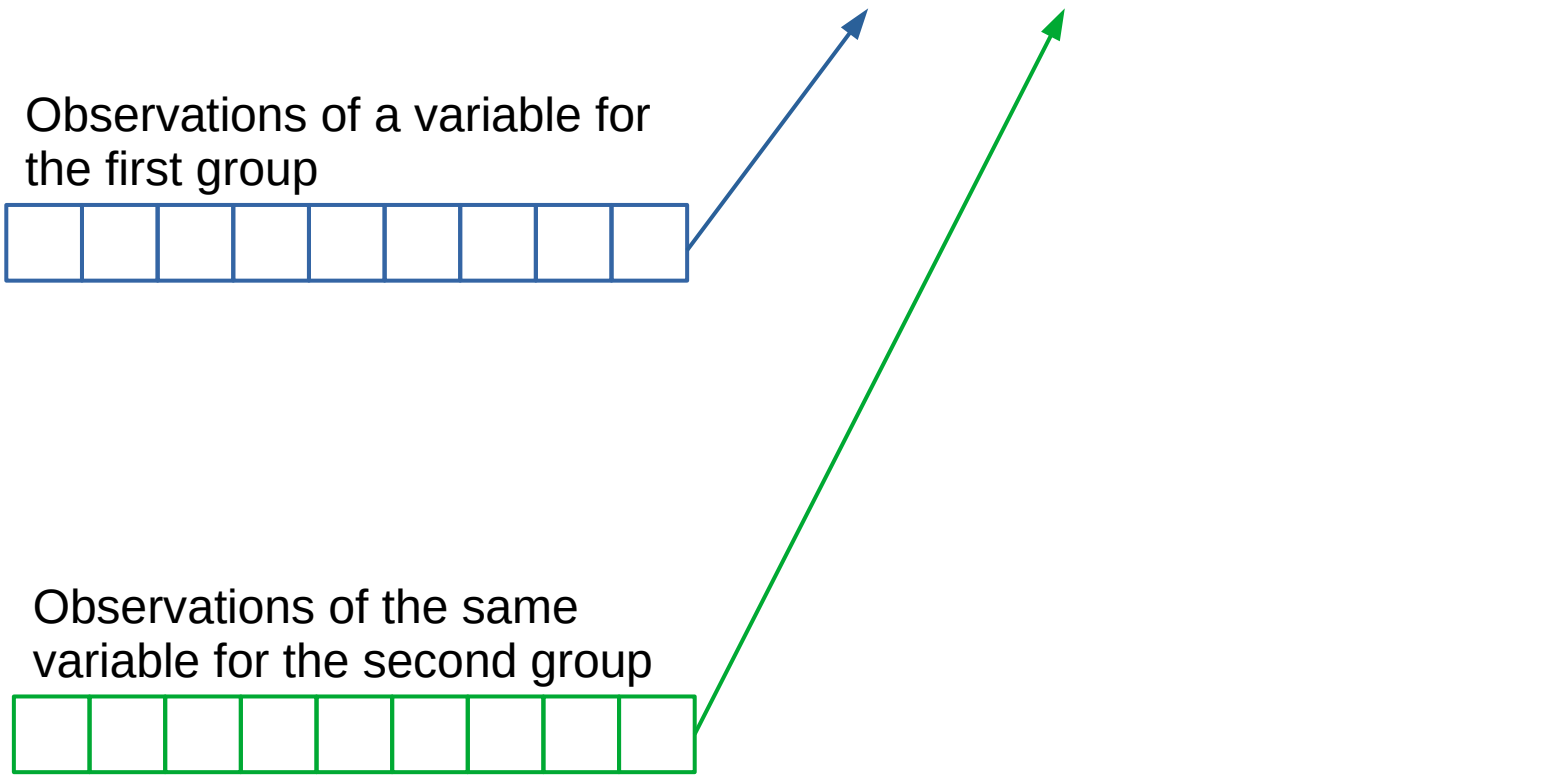


Has our data a normal distribution?

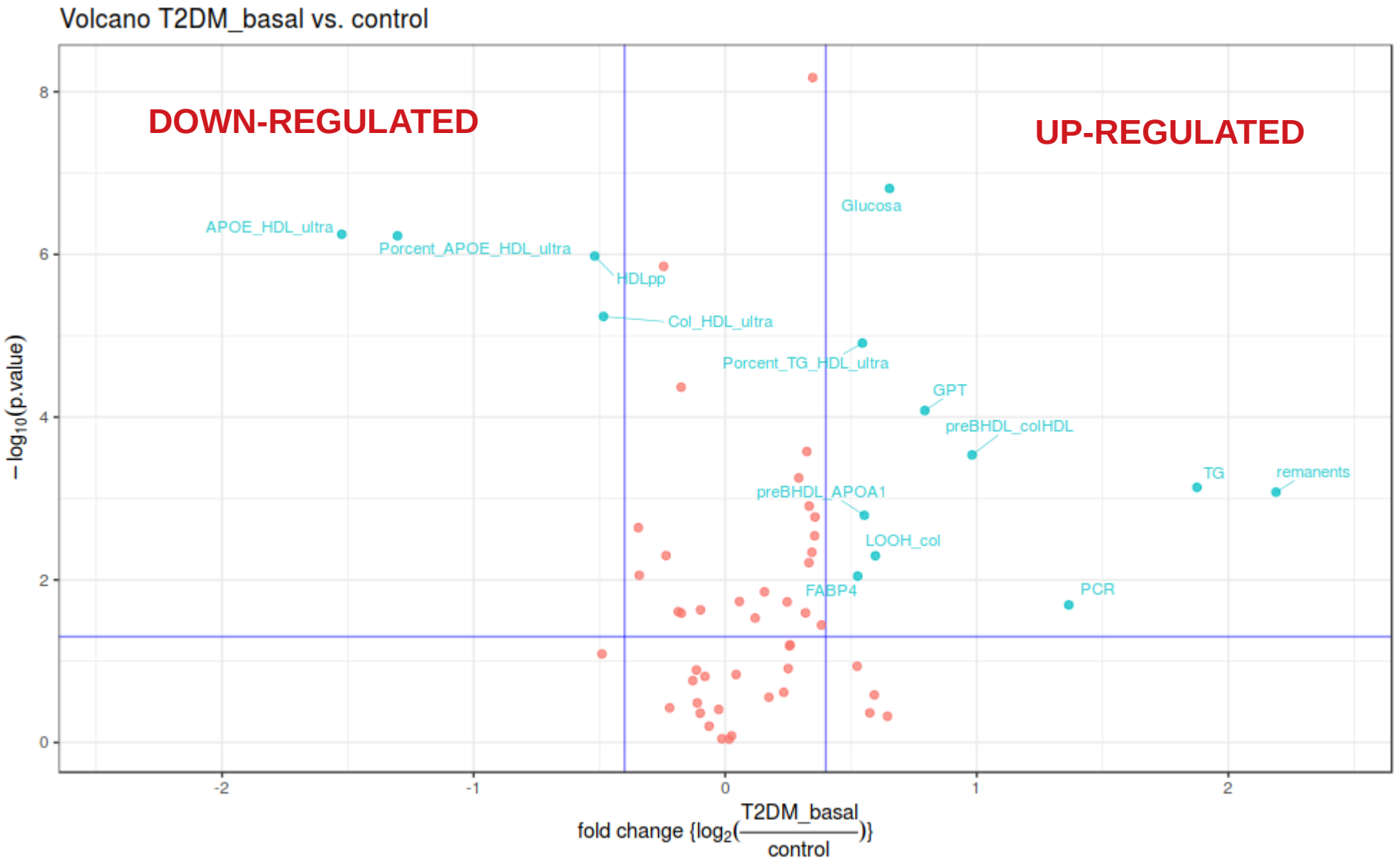


Kruskal–Wallis H test

```
pVal <- kruskal.test(list( var1, var2 ) )$p.value
```



Volcano plot



**p.values were calculated using the Student t test*